Graduate Research Plan

Research Title: Developing doped cryogenic rare-gas crystals as an atomic solid-state clock platform.

Introduction and Motivation

Atomic clocks are a crucial part of modern science and society. As the most sensitive clocks in existence, they provide the precise timing needed for GPS navigation, telecommunications networks, and global time standards. They also enable fundamental physics tests of general relativity and searches for physics beyond the Standard Model. At a basic level, atomic clocks work by repeatedly probing a well-defined atomic transition and comparing its frequency to that of a local oscillator. By stabilizing a local oscillator to an atomic transition, one realizes an extremely stable and accurate frequency reference. For this, one needs an atomic system with two well defined states separated by a known frequency [1]. Transitions between the hyperfine ground states of alkali metals, probed with microwave pulses, enabled a historic leap in clock stability and are still used as the time standard today [2].

Recently, coherent control over microwave transitions between the hyperfine ground states in a doped cryogenic rare-gas crystal has been demonstrated [3]. This finally opens the possibility of using these crystals and this newly demonstrated control to develop an atomic clock. Doped cryogenic rare-gas crystals offer a promising platform for both atomic clocks and precision measurement as they provide an optically transparent 'artificial vacuum,' which can confine a vast number of atoms or molecules exceeding ~10^{16} atoms per cubic cm; at the same time, weak interactions with the host matrix allow these dopants to retain many of their favorable gas-phase properties [4]. By hosting significantly more atoms or molecules than standard traps, one greatly increases the sensitivity possible. For example, an atomic clock's maximum sensitivity can be described using the Allan deviation, $\sigma(\tau) = \frac{1}{\nu 0*(2*T2)}\sqrt{\frac{Tc}{N*\tau}}$. Current state of the art microwave atomic clocks have allan deviation close to ~$1.5 * 10^{-13}/\sqrt{\tau}$ [2']. For a back-of-the-envelope comparison, if one grows a cubic scale doped crystal and can gain similar coherent control as demonstrated by Kanagin, with some other caveats to be discussed later, one could get $\sigma(\tau) = \frac{1}{\nu 0*(2*T2)}\sqrt{\frac{Tc}{N*\tau}} = \frac{1}{1.771\ GHz*(2*0.922\ ms)}\sqrt{\frac{5\ ms}{3*10^{16}*\tau}} \approx 10^{-16}/\sqrt{\tau}$. Realizing this potential, however, is nontrivial. It will require creating a well prepared initial state, performing a modified Ramsey style interrogation to mitigate inhomogeneous broadening common in these crystals, and implementing a way to readout the final state populations and locking our local oscillator to the atomic transition.

Research Plan

First, I will develop control over a well-defined hyperfine 'clock' transition in an alkali-doped rare-gas crystal and quantify its stability. Building off the work Kanagin has demonstrated, I will likely start with sodium-doped cryogenic neon crystals, co-depositing the species on a cryostat growth substrate. I will use microwave and RF pulse sequences to pump populations into the mF = 0 Zeeman sub-levels: a "clock" transition which has first-order insensitivity to magnetic fields. Control and pumping in these ground hyperfine Zeeman sublevels has been demonstrated and will be a good starting point [5]. From here I will characterize both inhomogeneous and homogenous spin relaxation coherence times, T2* and T2 respectively. Understanding these are important as they set how sensitive measurements of these populations can be and therefore the ultimate clock sensitivity. Comparing coherence times to existing measurements will be a useful check.

A common method for measuring phase shifts in frequency between an atomic transition and a local oscillator involves a method called Ramsey interferometry, and the sensitivity of these measurements is limited by the T2* coherence time. In doped cryogenic crystals inhomogeneous broadening results in incredibly short T2* times compared to their T2 [5]. T2* times are micro seconds compared to their greater than ms T2 times. To address this problem, a modified Ramsey sequence will be used whose stability is closer to the T2. The Hahn-Ramsey and similar modified Ramsey sequences have

been used in optical atomic clocks to help deal with inhomogeneities in laser probe fields, and they have also been proposed as a method to greatly reduce the effect of inhomogeneous broadening in doped solids [6][7]. Kanagin demonstrated a T2 time of 0.922 ms using Hahn spin echo, and I aim to approach this same coherence using modified Hahn-ramsey [3].

With this improved coherence and sensitivity provided by Hahn-Ramsey, interrogation on the doped crystal will provide sensitive measurements of the frequency shifts between the local oscillator and atomic transition periodically. The error signal between these two frequencies will then be used to adjust our local oscillator and lock it to the atomic hyperfine transition, creating a stable frequency reference. In a realistic first implementation of this setup one will use commercial quartz oscillators similar to those in other microwave atomic clocks, which will likely limit our sensitivity to only $\sim 10^{-13}/\sqrt{\tau}$, setting a reasonable first goal for our alkali doped cryogenic rare-gas crystal atomic clock [8].

Intellectual Merit

This project will explore whether a doped cryogenic rare-gas crystal can function as a microwave clock reference. Through this process, we will develop a deeper understanding of control over these hyperfine ground states, which is crucial for future improvements in doped crystals as a platform for quantum sensing and table top fundamental physics. Due to enhanced sensitivity via large dopant number, and the demonstrated potential for much longer coherence, doped cryogenic crystals are a promising platform for the next generation of frequency references. To sustain advances in precision measurement, quantum metrology, and quantum information infrastructure, increasingly precise frequency references are necessary. The coherent control of the hyperfine ground transitions developed in this project will have direct applications in using these crystals in quantum information science as these crystals have been proposed as platforms for quantum simulators, hybrid quantum systems, and quantum sensing, which perform well in these well defined two state system transitions we will be probing [3][1]. They also have already demonstrated their capability as nano-scale magnetometers, which will be crucial for higher resolution tomography and spectroscopy in chemistry and biology [5].

Broader Impact

Atomic clocks have become crucial in modern society as a time reference, and developing a solid-state atomic clock using doped cryogenic rare-gas crystals provides a new platform for time-keeping which is synergistic with quantum technological development on many fronts. The interdisciplinary nature of this research could enable reaching applications across fundamental physics, quantum information science, chemistry, and biology. Quantum metrology is also the bedrock of a system of standards on which industry evolves, so as this advances more precise time keeping will be necessary. It is important to note that doped cryogenic crystals have vast potential and sweeping impact as table top experiments. In addition to benefiting society through technological development, they can serve as a potent training ground for the next generation of scientists who could go into any of the aforementioned fields. As I have mentioned in my personal statement, I am dedicated to mentorship and participating in collaborative environments, and will use doped cryogenic crystals to facilitate this.

References:

1. C. L. Degen et al., Rev. Mod. Phys. 89, 035002 (2017)
2. Vladislav Gerginov at al., Metrologia 62, 035002 (2025)
3. Kanagin et al., arXiv:2508.21651 (2025)
4. J. D. Weinstein et al., Phys. Rev. Lett. 117, 175301 (2016)
5. U. Dargyte et al., Phys. Rev. A 104, 032611 (2021)
6. T. Zanon-Willette et al., J Phys.: Conf. Ser. 723, 012057 (2016)
7. N. V. Vitanov et al., Phys. Rev. A 91, 033406 (2015)
8. K. Syzmaniec et al., J. Phys.: Conf. Ser. 723, 012003 (2016)
9. A. D. Ludlow et al., Rev. Mod. Phys. 87, 637 (2015)